UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

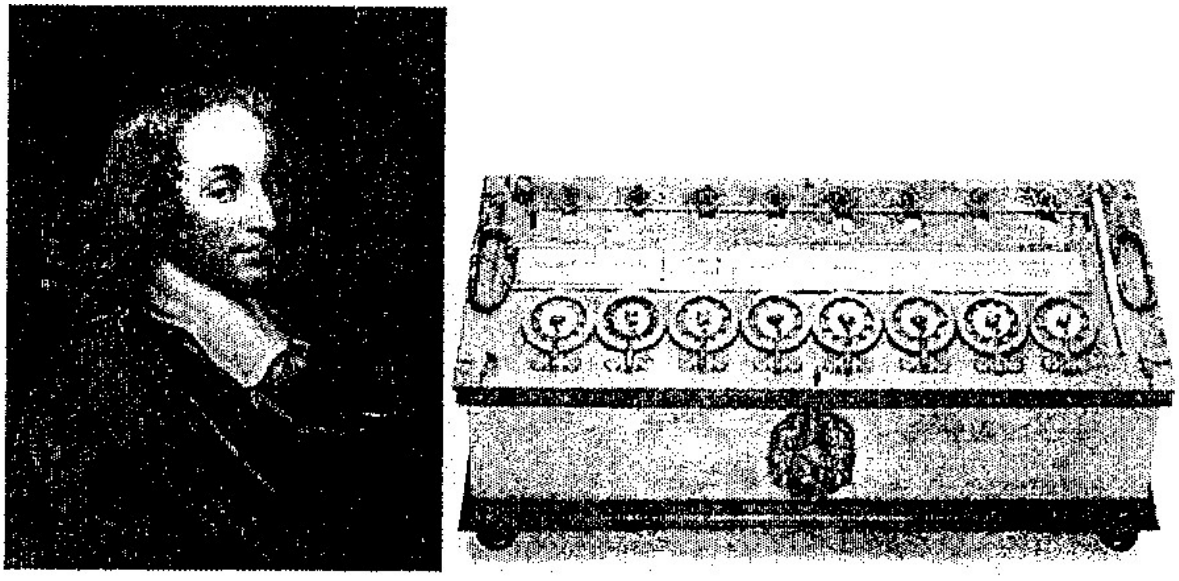
a) Identify the Von Neumann architecture input and output components based on the excerpts from the web.

The Jacquard Loom is a mechanical loom invented by Joseph Marie Jacquard in 1801, which simplifies the process of manufacturing textiles with complex patterns. The loom is controlled by punched cards with punched holes, each row corresponding to one design row. Multiple rows of holes are punched on each card, and the many cards that compose the textile design are strung together in an orderly manner.

b) Identify ONE (1) similarity and ONE (1) dissimilarity between the mechanical calculating machine and Atanasoff Berry Computer.

QUESTION A2 **(5 marks)**

Question 1 is based on the following Figure 1.



Inventor 'Y' was France's most celebrated mathematician and physicist. In 1642, inventor 'Y' invented and build the first working calculating machine as a means of helping his father to perform tedious tax accounting.

a) Who is inventor 'Y'?

b) Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.

c) Which programming language is named after inventor 'Y'?

d) State TWO (2) mathematical operationsthat can be performed byinventor Y's calculating machine.

QUESTION A3 **(5 marks)**

a) List THREE (3) interfaces to external device.

b) Discuss 'hot connection' of device.

QUESTION A4 **(5 marks)**

John Von Neumann is a mathematician who is one of the pioneers in computer advancement, and his contributions are still relevant to the modern computer.

a) Draw the Von Neuman architecture and give relevant examples each.

b) What is the name of the actual device that John Von Neumann himself developed?

QUESTION A5 **(10 marks)**

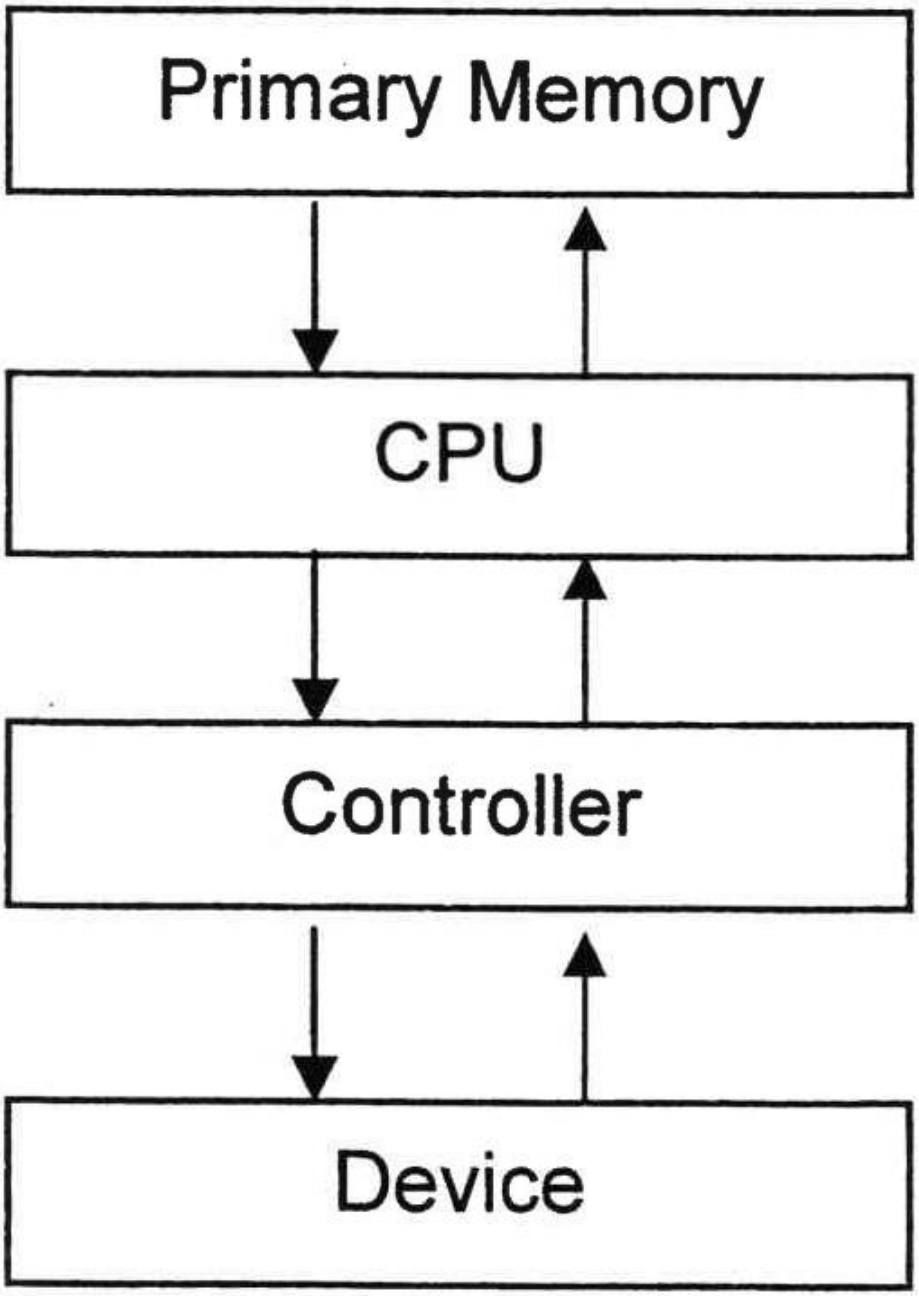
Input output technique can be segregated into programmed, interrupt-driven and direct memory access (DMA).

a) What is DMA?

b) How can the DMA technique improve the overall computer performance?

c) Discuss TWO (2) major drawbacks of programmed and interrupt-driven input output technique that inspire hardware developer to introduce DMA.

d) Conventional access is depicted in Figure 2. Redraw Figure 2 to illustrate DMA architecture.



QUESTION A6 **(10 marks)**

a) Complete the taxonomy of parallel processor architecture illustrated in Figure 4.

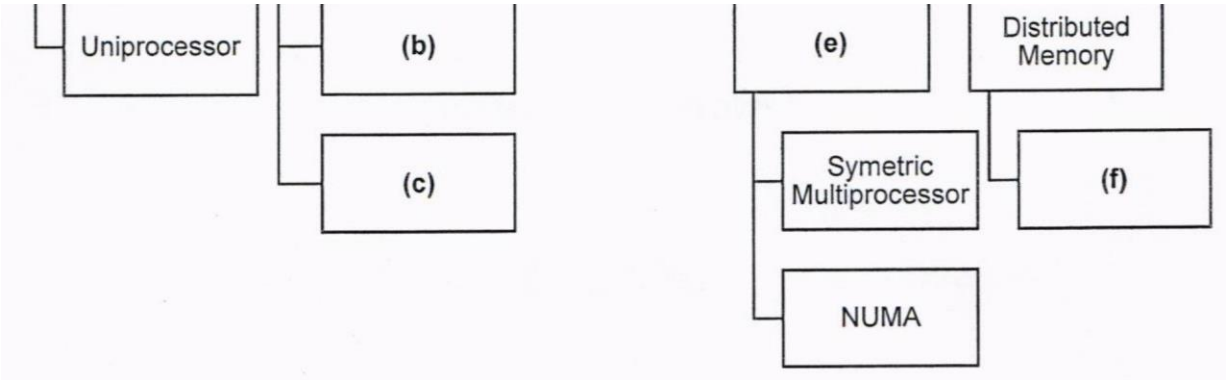
Processor Organization

Multiple Instruction Single instruction

(d)

(a) Multiple Data (MIMD) Multiple Data (SIMD)

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(e)

(b) I\ilemory

b) Discuss TWO (2) disadvantages of symmetric multiprocessor (SMP).

QUESTION A7 **(10 marks)**

a) Convert -176.812510 to floating point number in IEEE single precision format in hexadecimal value form.

b) Convert 3FOOC00016 in IEEE single precision floating point number to decimal value form.

QUESTION A8 **(10 marks)**

Describe FOUR (4) type of access methods based on hardware implementation of the storage device. Provide ONE (1) appropriate example for each access type.

QUESTION A9 **(10 marks)**

a) Draw Single Instruction, Single data (SISD) architecture with the following information: - CU is the control unit - PU is the processing unit - MU is the memory unit - IS is the instruction stream - DS is the data stream

b) List TWO (2) examples of shared memory Multiple Instruction Multiple Data (MIMD).

c) Discuss the difference between what happens if a virus attacks shared memory Multiple Instruction Multiple Data (MIMD) and distributed memory Multiple Instruction Multiple Data (MIMD).

d) Explain ONE (1) benefit of using distributed memory Multiple Instruction, Multiple Data

stream (MIMD) architecture with example.

QUESTION A10 **(10 marks)**

a) Define interrupt.

b) Describe TWO (2) different types of interrupt.

c) Discuss ONE (1) advantage and ONE (1) disadvantage of interrupt

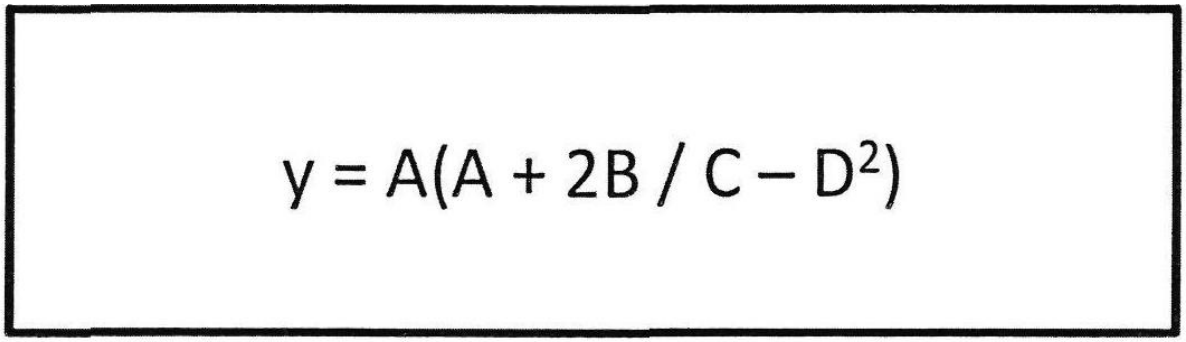
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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.



Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.

Zero-address	One-address	Three-address
PUSHp	LOADp	ADD (p,q,r)
POPp	STORp	SUB (p,q,r)
ADD	ADDp	MUL (p.q.r)
SUB	SUBp	DIV(p,q,r)
MUL	MULp	
DIV	DIVp	